Development of a GLOF Warning System through Strong Endurance Drone-based Long-Range (LoRa) Communication for the Upper Catchment of the Tista River

(Proposal under NIDM Research Project on Disaster Management: Innovation & Action Research)

Presented by

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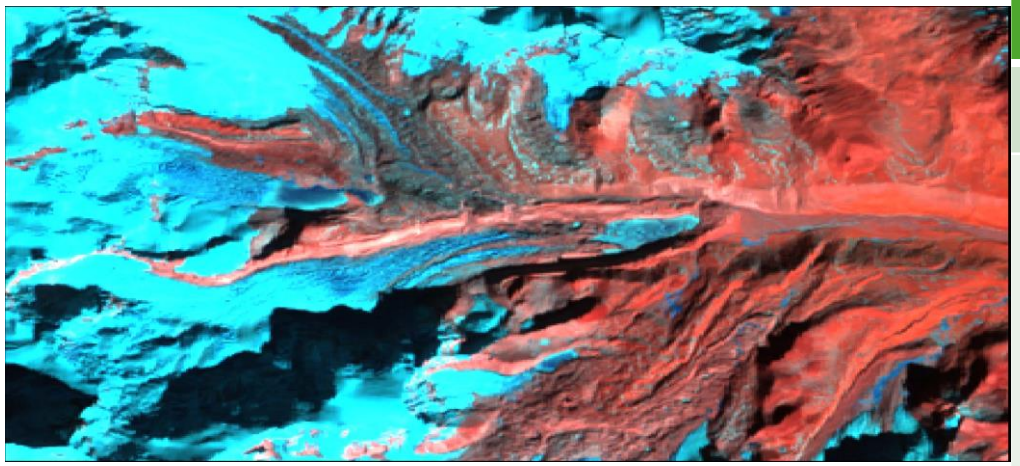
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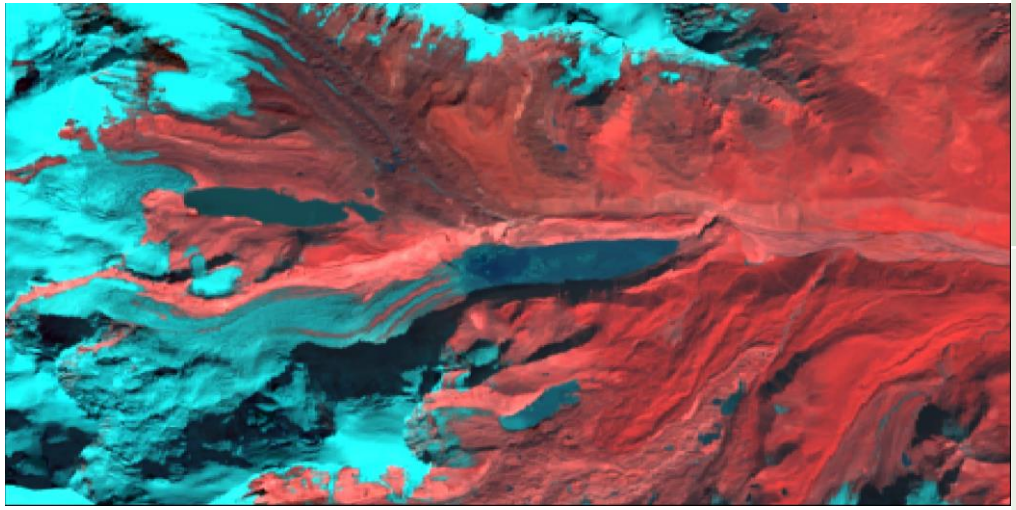
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Problem and Objective

Upper Tista Basin, Sikkim [Dec 1990]



Upper Tista Basin, Sikkim [Dec 2023]

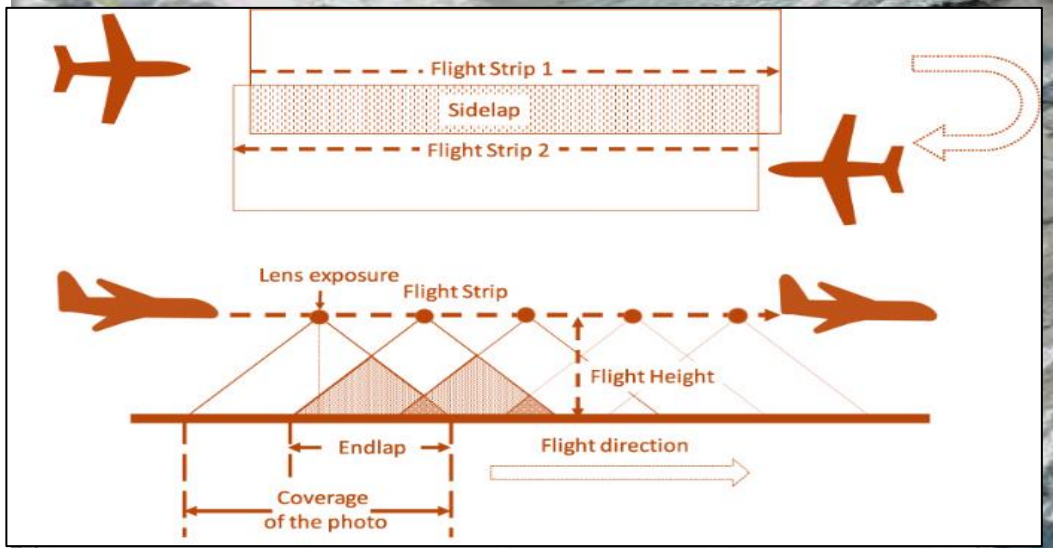


Glacier Lake Outburst Flood (GLOF)	
Problem	**Objective**
1. Limitations of existing imagery and barrier of Himalayan region - Low spatial resolution (6-10 m per pixel) - Bad weather condition - Rugged terrain, unavoidable wind speed - Ground based system not suitable - Private satellite images are very costly (10-30$/$Km^2$)	**1. Development of a fixed wing, high endurance drone or UAV** **2. Investigating and communicating ice fracturing and glacier lake outburst**
2. Conventional mathematical model (e.g. GIS/DEM) - Static based on elevation and slope - Glacier level change, movement not detected - Limited pattern recognition	**3. Developing an ML/AI technique to integrate and analyze from both satellite and drone images**
3. Data centric non-comprehensive Early Warning System (EWS) - Monitoring same location in same direction - Long time to predict - Lack of relevance in state and district level	**4. Implementation of a 4G/5G based warning system**

Glacial Lake Monitoring and Communication

Zanak, North District, Sikkim, India

Alternative Names:	Dzanak, Dzānak, Zanak
Type:	Populated place - a city, town, village, or other agglomeration of buildings where people live and work
Mindat.org Region:	Chungthang, North Sikkim district, Sikkim, India
Region:	North District, Sikkim, India
Latitude:	27° 54' 5" N
Longitude:	88° 16' 24" E
Lat/Long (dec):	27.90163,88.27356
Köppen climate type:	ET : Tundra

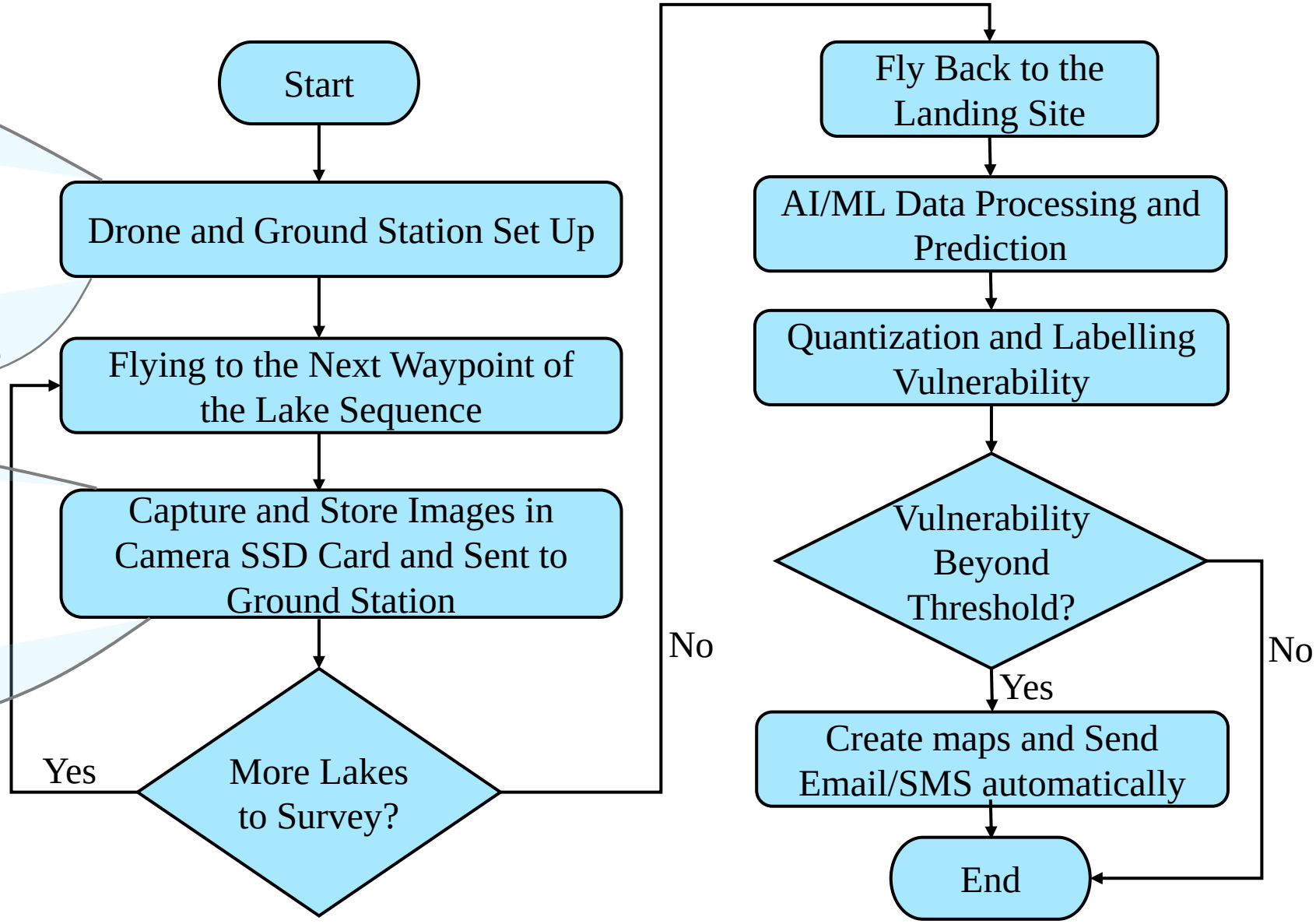


Lake 3
Lake 4
Lake 2
Lake 1
Drone trajectory over a lake
4G/5G connection to send instant images at regular intervals
Dzānak
Camp
LoRa Connection to control the drone (telemetry)
Image stored in SSD card

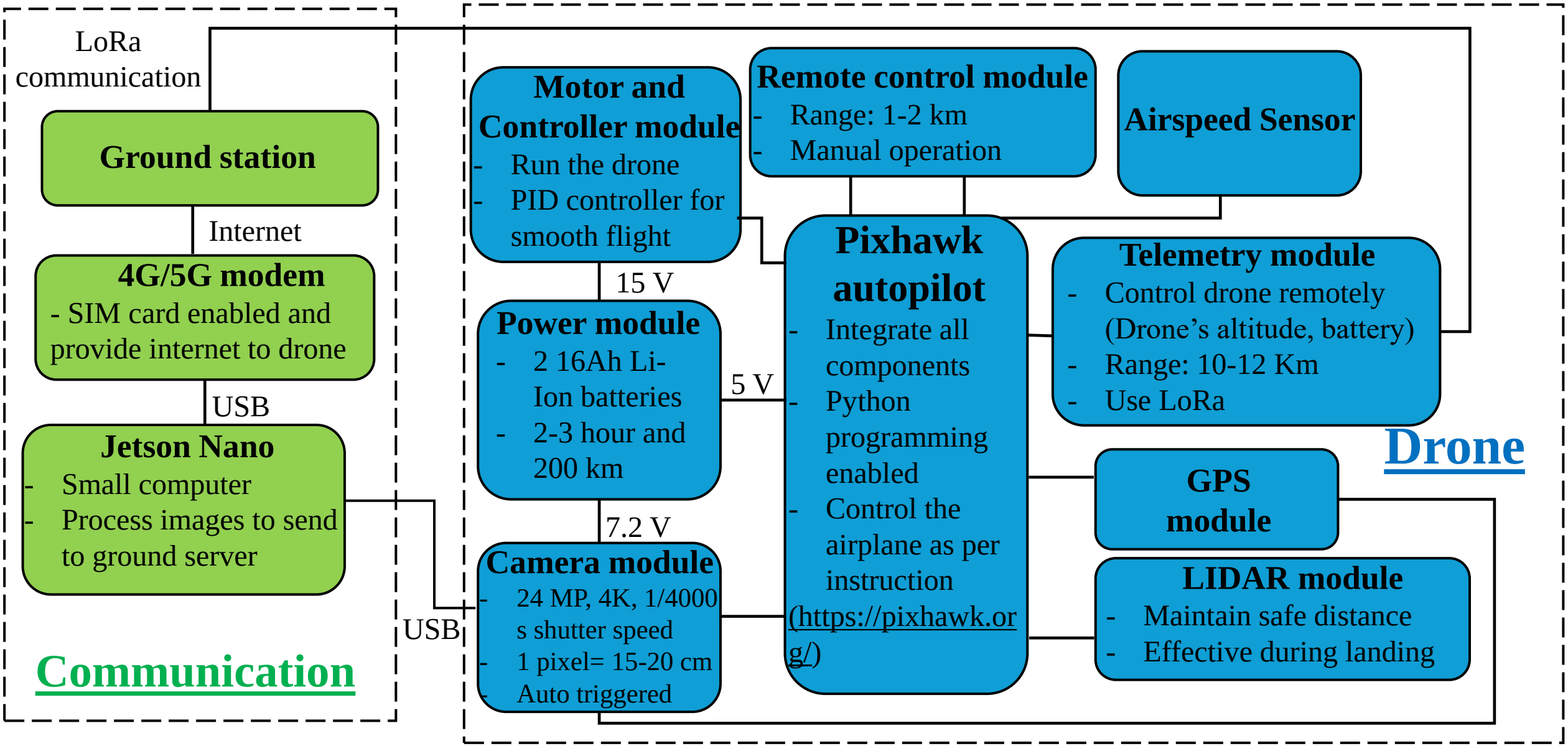
Overall Procedure

Mission Planning commands:
(**www.ardupilot.org/planner**)
- Python programming
- Ground clearance: 450-800 m
- Automatic take-off
- LoRa Setup
- Load waypoints of different glacial lakes

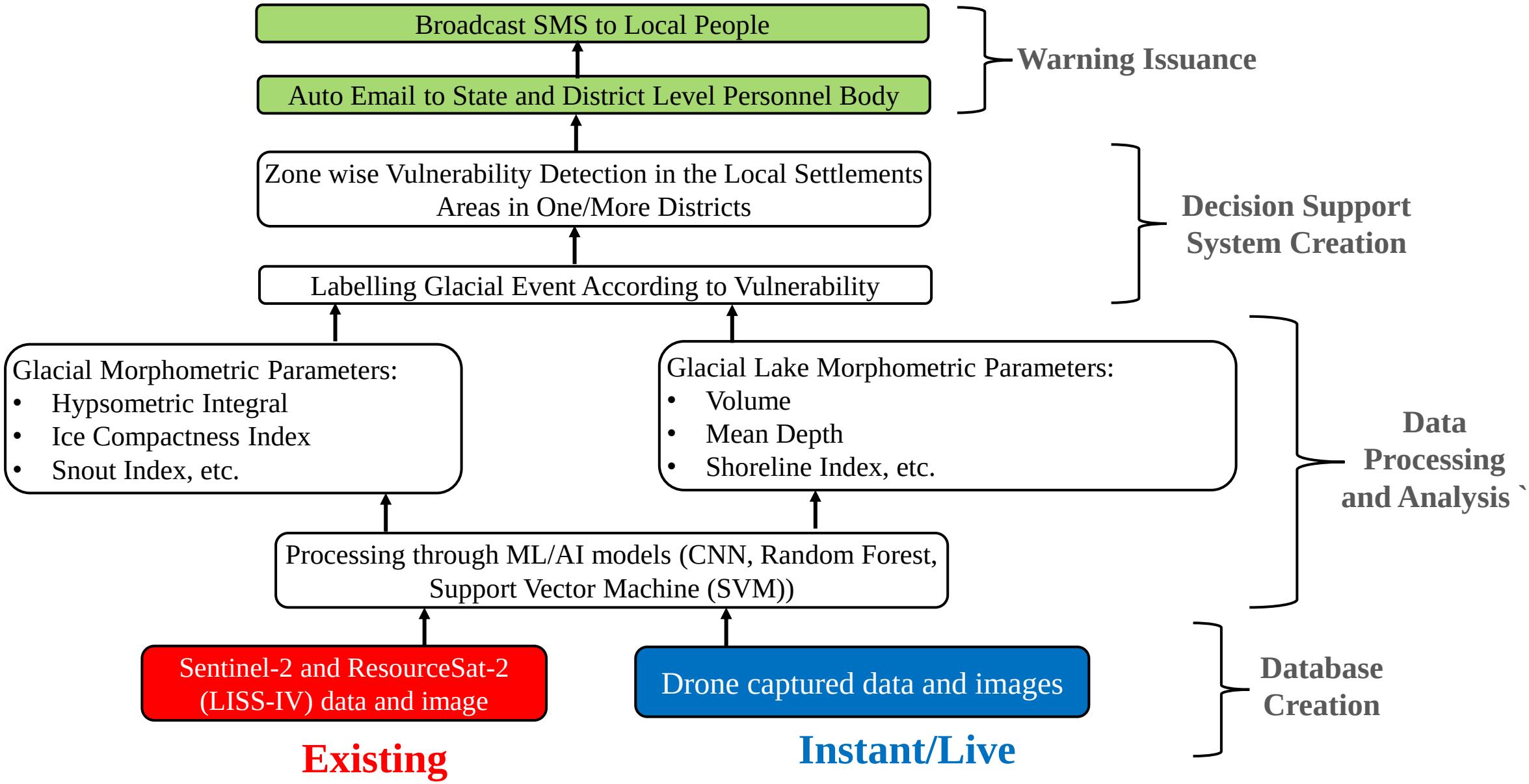
Image store and transmission:
- Co-ordinate of each image from GNSS receiver stored and sent (**www.rtklib.com/rtklib.htm**)
- Camera triggering in each 60 m distance



Drone Configuration and Communication



Early Warning System



Relevance for State and District

- Zero casualties through DRR as per disaster management act
- Determination of level of vulnerability and issue a warning at local and district level
  - Communication through Email, SMS and Website update to proper Personnel/Body
- Forecasting of GLOF for next few days and even weeks
- Proposed to conduct GLOF awareness workshop at state level with the help of SRM-SIKKIM campus

Work Plan

ACTIVITIES	I YEAR				II YEAR			
	3	6	9	12	3	6	9	12
1. Student selection (UG level), Equipment purchase, Satellite data acquisition								
2. Field visit and data collection: Drone image collection, Database creation, Ground based evidence								
3. Image processing through software: Object detection and image classification								
4. Glacial and glacial lake Morphometric analysis and quantification								
4. ML based model and decision Support System								
5. Early Warning System for GLOF: Monitoring of Dynamic changes of glacial lake, Checking warning at various level, Message broadcasting using 4G/5G network								
6. Report Submission								
7. Research publications and patent work								

Proposed Budget

A. Equipments			**Price (Rs.)**	**Weight (gm.)**
1	Fixed wing body	X8 Skywalker 300	180000	1000
2	Motor	Scorpio SII-4020-420 KV	16000	288
3	Controler	Castle Phoenix Edge 50A regler	13000	130
4	Servo motors	2 × DES 678 BB MG servo	10260	80
5	Propellers	Aeronaut CAMcarbon 17”x9”	2000	40
6	Autopilot	Pixhawk HKPILOT 32	12000	40
7	GPS	UBLOX NEO-M8N GPS	2500	23
8	Telemetry	HKPILOT 500MW Telemetry	9512	50
9	Air speed sensor	HKPILOT 32 digital air speed	5400	10
10	Lidar	Lightware SF10/A	28000	20
11	RC receiver-sender	Futuba T14SG-R7008SB	33735	50
12	Battery	2 × 4S5P 16Ah Li-Ion (Panasonic NCR18650B cell)	40000	2000
13	Camera	Sony Alpha ILCE 6000L with 16-50 lens	45000	400
14	Camera trigger release	Sony Alpha6000 trigger cable	4000	10
15	GNSS receiver	Emlid Reach (base and rover)	60000	35
16	Hot shoe cable	Tuffwing/Reach hot shoe cable	8540	20
17	4G Modem	LTE Wireless dongle	2000	40
18	Jetson Nano Computer	NVIDIA Jetson Nano Developer Kit	26000	250
19	External Memory	External SSD 128GB	2000	60
20	USB to Micro-USB	Micro USB Cable 2 A 1 m UMC100	300	30
21	SIM Card	Any 4G LTE SIM card with net pack	600	5
22	Insurance	Drone Insurance through TROPOGO	50000	
		GST	99152.46	
		Total (Equipments)	649999.46	**4581**
B. Contingencies			10000	
C. Student Internship			20000	
D. Field Work Real-time Data Collection			106000	
Total cost			**785999.46**	
Overhead (10%)			13600	
Grand Total			**799599.46**	**~8 Lakhs**

Deliverables and Outcome

- **A low-cost high endurance drone**
  - Takeoff and landing in steep terrain
  - Sustain against strong katabatic winds
  - No unreliable compass used
  - Tolerate battery deterioration in low temperature, limited technical support
  - Easy drone launch and landing process
- **An efficient Machine Learning model, determining GLOF**
  - Much better result than conventional mathematical model
  - No dependency on satellite and ground system images
- **Instant and automatic EWS system**
- **UG student** internship supported and **GLOF awareness program**, organised through **SRM Sikkim** campus
- Project can be extended for land slide and other disaster scenarios

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Thank You!!